Executive BI Report

Daily Campaign Budget Analysis

March 17th, 2026

Period: 02-01-2025 to 16-12-2025

10,380 daily records | 100 orders | 5 Customers

Technical Round Submission

Factor 11 GmbH

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1. Executive Summary

- This report analyses 10,380 daily budget orders across 100 advertising orders for 5 unique customers (Audi, BMW, Mercedes, Porsche, Volkswagen).
- The duration is from January 2025 to December 2025.
- Total allocated budget is €: 2,820,136 across 5 product families.

HEADLINE FINDINGS:

- **The campaigns experience a dramatic budget fall beginning in July 2025.**
- **Monthly budget peaks at €: 455,898 in June before dropping 99% down to €: 3,110 by December.**
- **Active orders fell from 69 to 1.**

ROOT CAUSE:

- **No serious campaign renewal after June 2025.**

PROJECTION FAILURE:

- **At peak as per the average daily budget, project portfolio valued at €: 5.3 Million.**
- **Trend implies valuation at €: 67k**

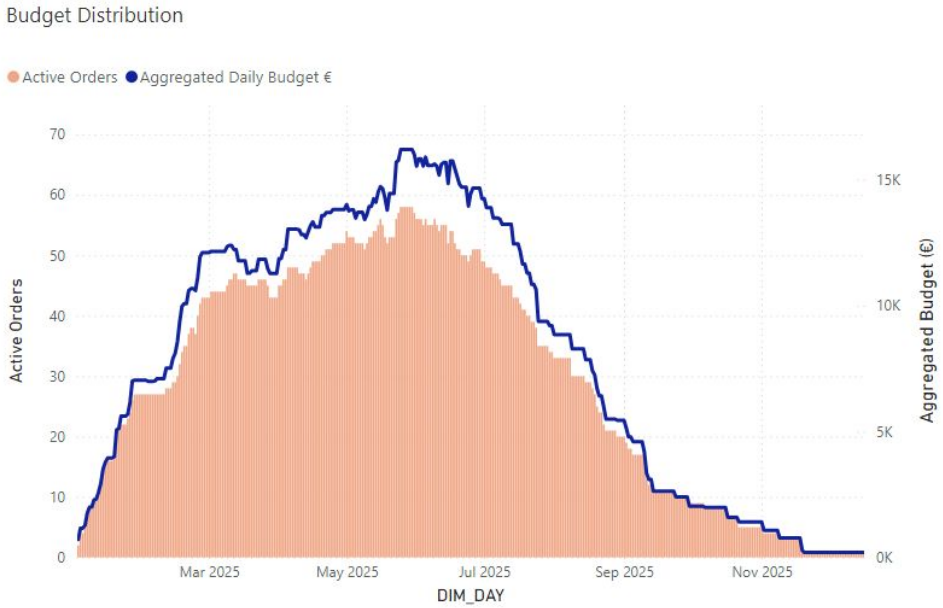
ADDITIONAL FINDINGS

- Porsche and VW concentration risk at around 25% of total budget (both exceeding the 20% threshold)
- 18 long-tail orders with low daily budgets.
- Product mix is relatively balanced across 5 families.
- Day of week budget allocation is statistically uniform.

2. Immediate Actions

Three immediate actions are recommended:

1) Emergency data pipeline review before Q1 2026 campaign renewals.
2) Customer diversification to reduce dependency on Porsche and Volkswagen.
3) Customer growth programmes for Audi and Mercedes.



- The Budget peaks in mid June.
- Active Orders peak in mid June.
- Negative cliff for both afterwards
- Both quantities follow a similar pattern.

3. Key Values

METRIC	VALUE
Date Range	2025-01-02 to 2025-12-16 (349 days)
Total Budget	€ 2,820,136
Peak Monthly Budget	€ 455,898 (2025-06)
Unique Orders	100
Unique Customers	5
Product Families	5
Average Daily Budget / Order	€ 272
Average Campaign Duration	104 days
June - December decline	- 99.3%

- All orders have fixed daily budget.
- Average Pacing Ratio is 1.0 (perfect spending over time)
- All Weekdays budgets are uniformly distributed.
- June and July have most active orders (69)

4. Priority Items

- **CRITICAL** (Budget cliff):
  - Daily budget fell 99.3% from €: 15,197 per day (June) to €: 194 per day (December).
  - 69 Active orders (June) to 1 Active order (December)
- **HIGH** (Campaign renewal gap):
  - Last new campaign started on 26-06-2025
  - Pipeline replenishing should be high priority
- **HIGH** (Budget concentration risk):
  - Porsche, VW, BMW cross the concentration threshold of 20%.
  - Dedicated account handling and retention strategy is recommended.
- **MEDIUM** (Low Budget and Long duration campaigns):
  - 18 orders run more than 60 days on budget below €: 149 per day. (Total: €: 186,119)
  - Evaluate operational cost versus return
- **MEDIUM** (Tenant Concentration):
  - “Digital Solutions Ltd” managed 24% of total budget. (Total: €: 682,319).
  - “Native Ads” are shorter duration with higher Average budget (€: 324).
- **LOW** (Data quality issue - “UNKNOWN” salesperson):
  - “UNKNOWN” salesperson accounts for 17% of Budget share. (‘UNKNOWN’ also in Sales Team and Department)
  - Possible Data Attribution Gap.

5. Dashboard Visuals and Tables

Visual: 1 (Bell Shaped Budget Curve)

- Peaked in June
- Massive drop in Q3, Q4
- Elaborates budget cliff
- Shows Active orders drop

5. Dashboard Visuals and Tables

Visual: 2 (Rise and Fall Pattern - Monthly Active Orders)

- Gradual increase until June
- Stagnant July
- Cliff in second half of 2025
- Shows Active orders drop

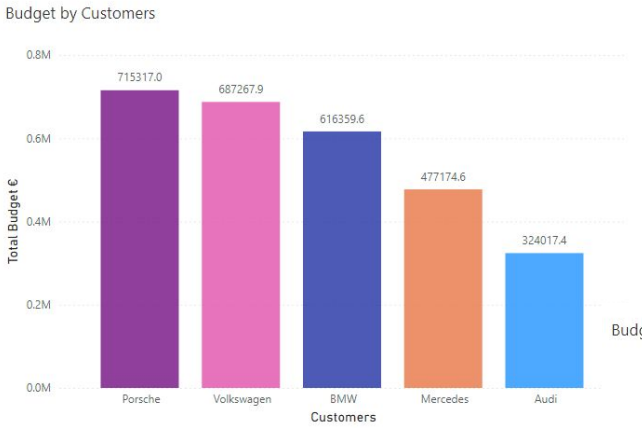
5. Dashboard Visuals and Tables

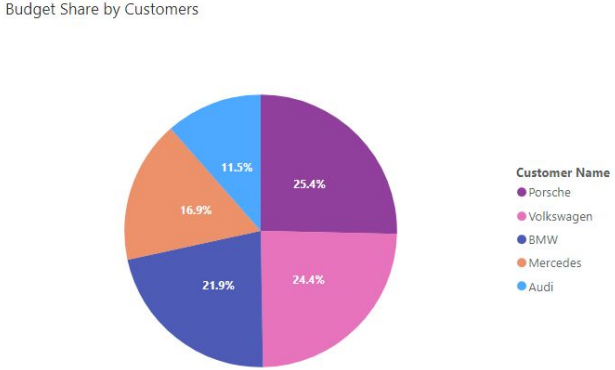
Visual: 3 (Frequency of Starting and Ending Orders per Month)

- Peak starts in January.
- Average 16 orders/month until June.
- Average 9 orders ending per month, February onwards.
- No new orders from July onwards.
- This explains the sharp decline in monthly active orders in second half of 2025.

5. Dashboard Visuals and Tables

Visual: 4 (Unique Customers and Budget Share)



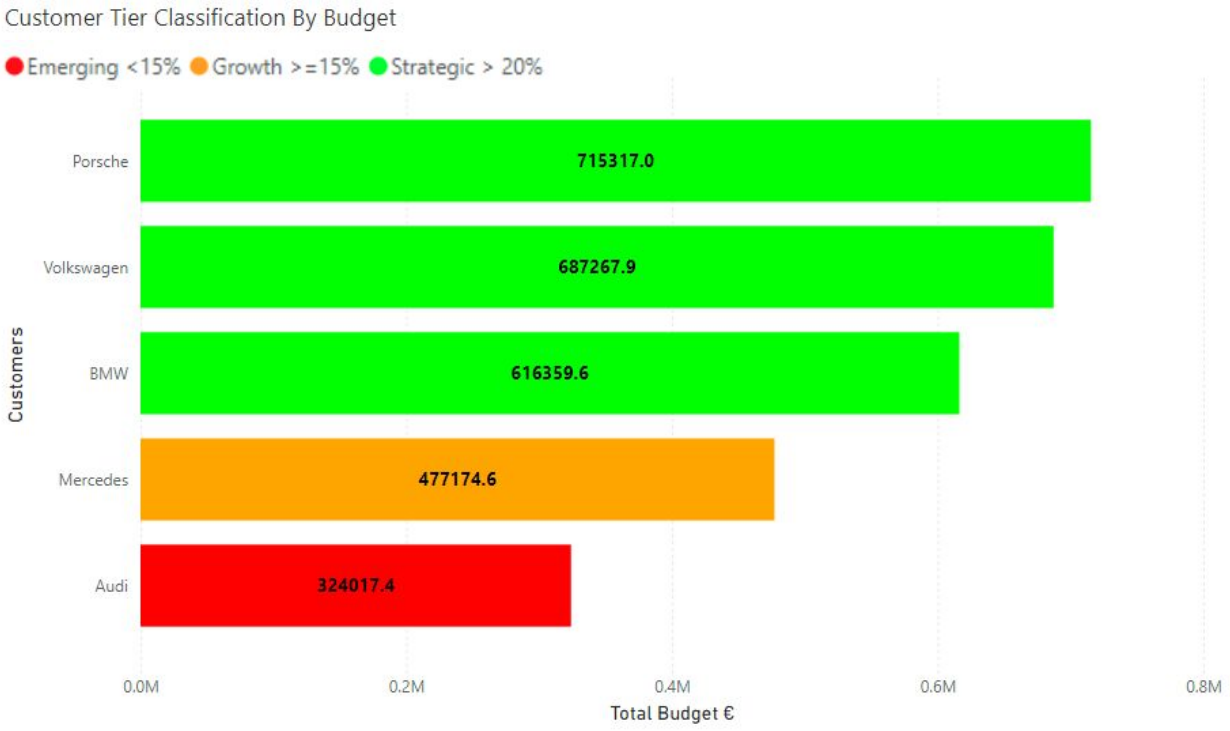


- Porsche with max budget.
- VW close second.

- Porsche and VW control ~50% of the budget.
- Losing them means half of the portfolio gone.
- Porsche and VW cross the maximum concentration threshold of 20%.

5. Dashboard Visuals and Tables

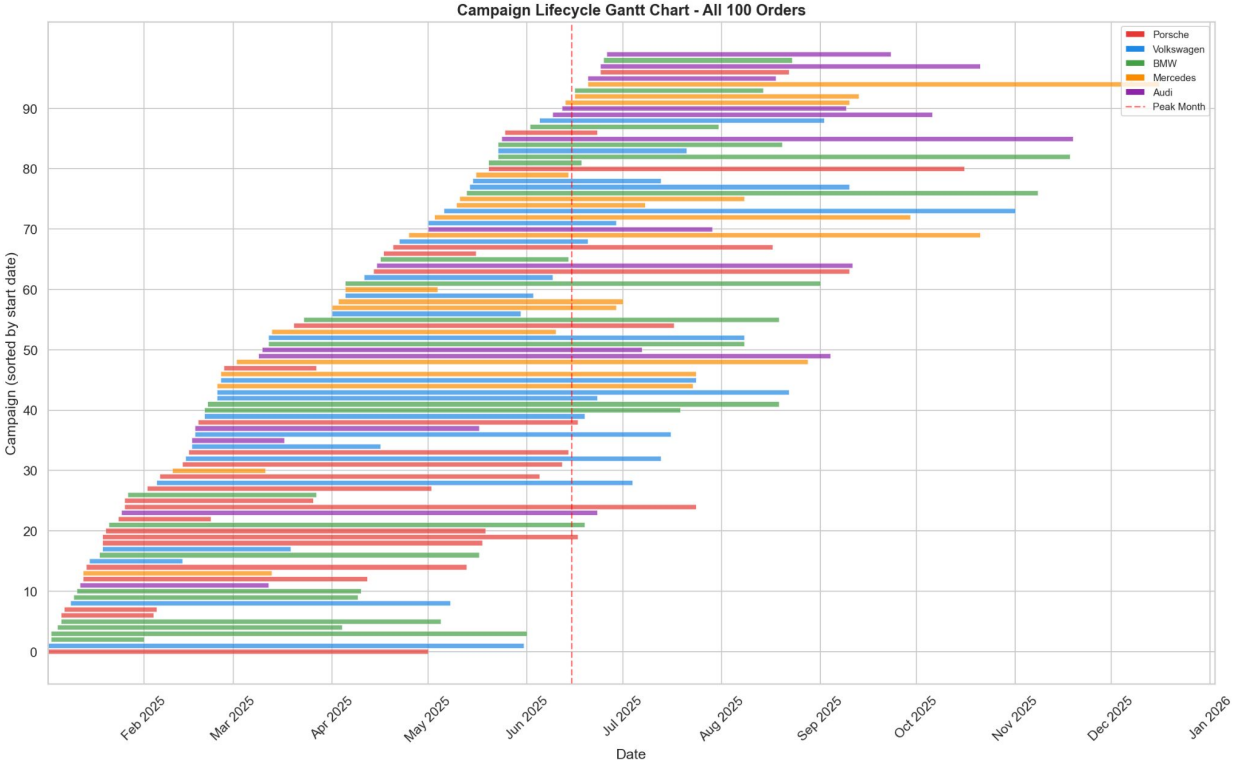
Visual: 5 (Customers Tier Classification)



- Strategic or Tier A customer is the one with more than 20% of the budget share.
- Porsche and Volkswagen are Strategic.
- BMW is borderline strategic.
- Mercedes and Audi under-indexed, should be considered under growth programs.

5. Dashboard Visuals and Tables

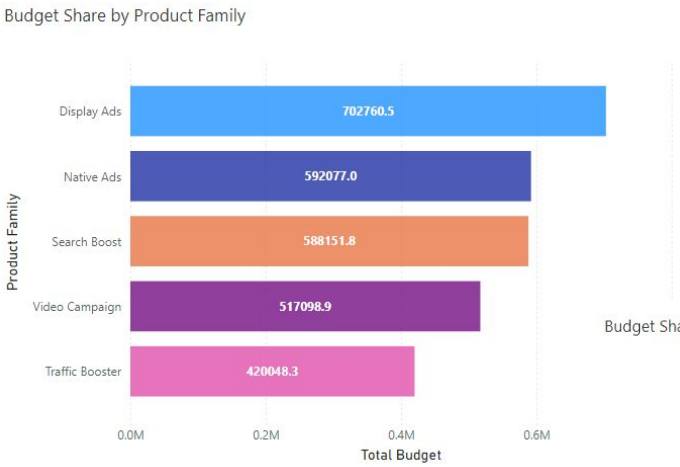
Visual: 6 (Campaign Lifecycle for all Customers)

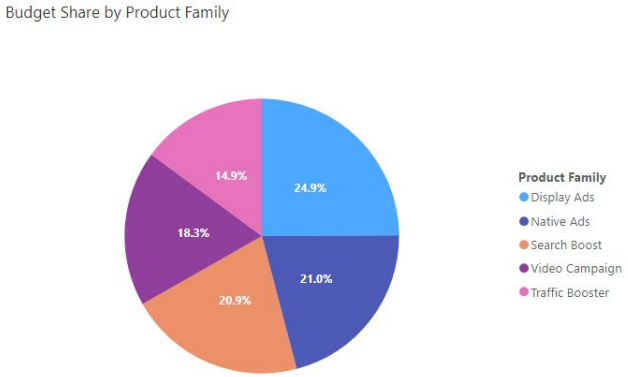


- Most campaigns starting in first half of 2025
- Highest running frequency in July
- Second half is dominated by campaign ends
- 69 campaigns from Porsche, VW, BMW
- Average campaign duration is 104 days.

5. Dashboard Visuals and Tables

Visual: 7 (Product Families and Budget Share)

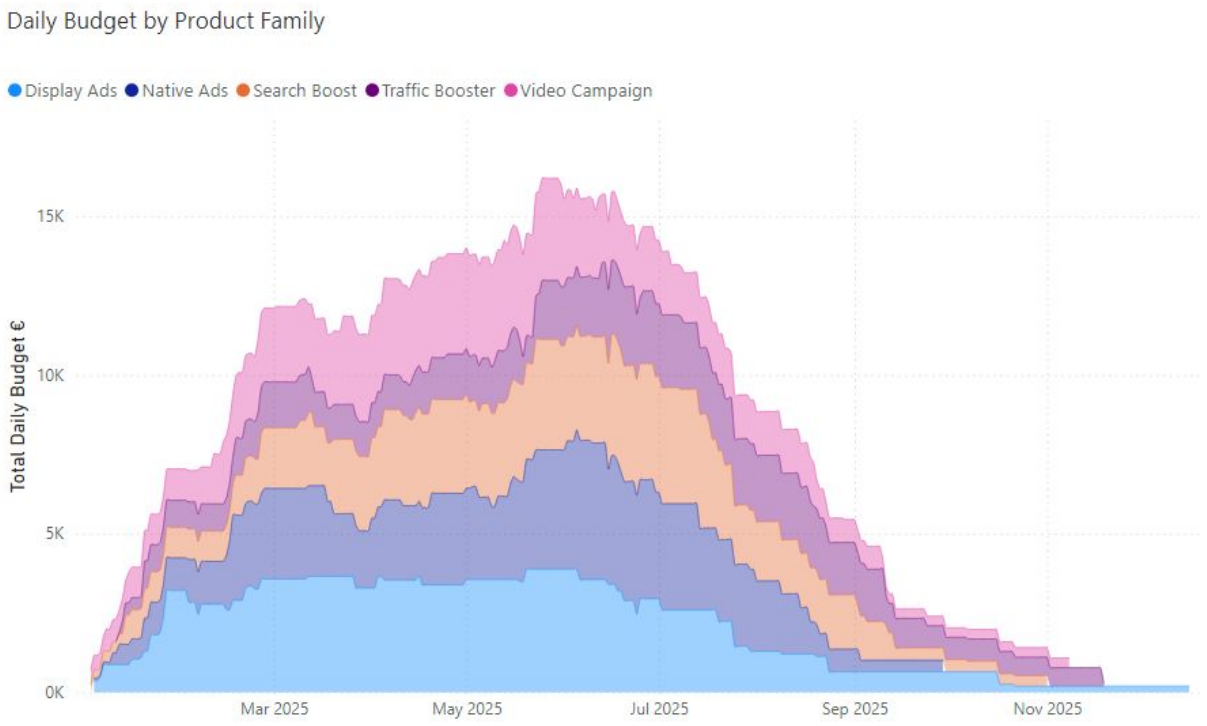




- Display Ads dominates the product family budget share.
- Native Ads and Search boost weigh the same.
- Display Ads controls quarter of the budget.
- Display Ads crosses the maximum concentration threshold of 20%.

5. Dashboard Visuals and Tables

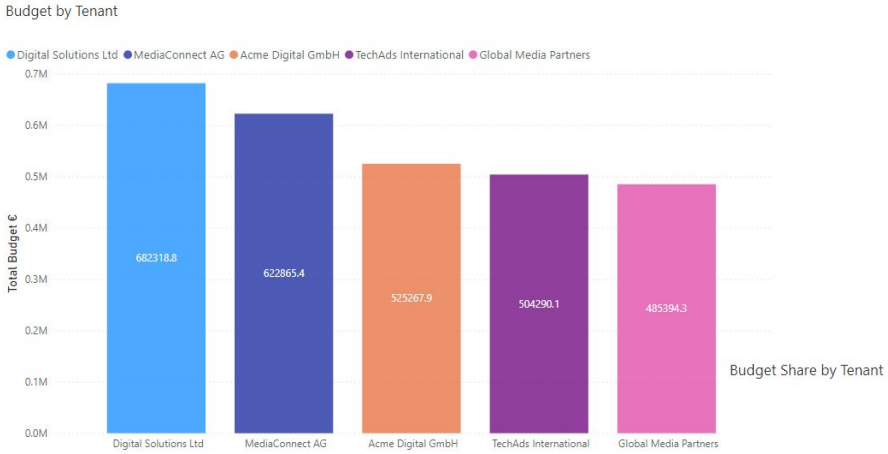
Visual: 8 (Product Family Daily Budget Distribution)

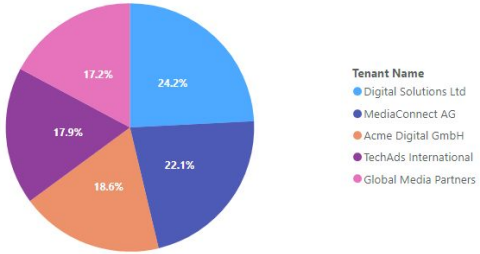


- All product families peak in June as well.
- Product spends also form normal distribution bell curve.
- They also follow same peaks and falls as active orders and budget.

5. Dashboard Visuals and Tables

Visual: 9 (Tenant Categorization and Budget Share)

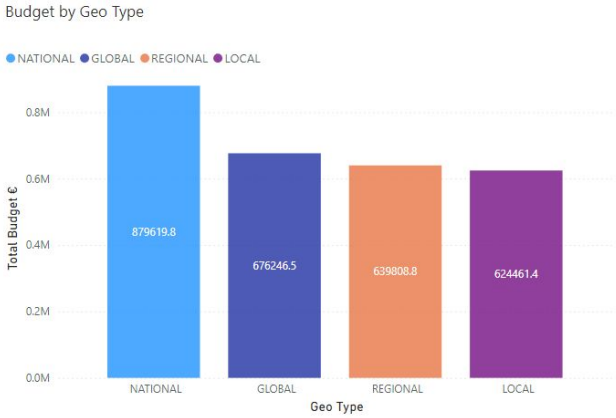




- Digital Solutions Ltd. takes most of the budget.
- Media Connect AG is a close second.
- Digital Solutions Ltd. and Media Connect AG cross the maximum concentration threshold.

5. Dashboard Visuals and Tables

Visual: 10 (Geographical Categorization and Budget Share)



- UK is a dominant market by a huge margin.

- National Targeting leads the budget share.

5. Dashboard Visuals and Tables

Visual: 11 (Billing Types Budget Share Analysis)

Billing Type	Net Budget	Total Orders	Av. Daily Budget	% Budget Share
Traffic	€ 922,522	29	€ 307	32.7 %
CPM	€ 690,957	24	€ 271	24.5 %
CPC	€ 604,001	24	€ 239	21.4 %
Fixed	€ 602,655	23	€ 260	21.4 %

- Billing types have different budget levels.
- Traffic type dominates with Net budget and Average daily budget.

Test	Comparison	Statistic	P-Value	Significance	Interpretation
Kruskal-Wallis	Billing Types	H = 309.5678	0.0000001	Yes	Significant difference

- Significance test tells that categories are different.
- Billing type mixed strategies matter for campaigns.

5. Dashboard Visuals and Tables

Visual: 12 (Product Family Budget Share Analysis)

Product Family	Net Budget	Total Orders	Av. Daily Budget	% Budget Share
Display Ads	€ 702,760	24	€ 272	24.9 %
Native Ads	€ 592,077	23	€ 323	21.0 %
Search Boost	€ 588,152	20	€ 242	20.9 %
Video Campaign	€ 517,099	19	€ 261	18.3 %
Traffic Booster	€ 420,048	14	€ 269	14.9 %

- Product families have different budget levels too.
- Display Ads have biggest share in Net Budget but not highest Average daily budget.

Test	Comparison	Statistic	P-Value	Significance	Interpretation
Kruskal-Wallis	Product Family	H = 343.6204	0.0000001	Yes	Significant difference

- Significance test tells that product families are different.
- Product family mixed strategies matter for campaigns.

5. Dashboard Visuals and Tables

Visual: 13 (Days of the Week Budget Share Analysis)

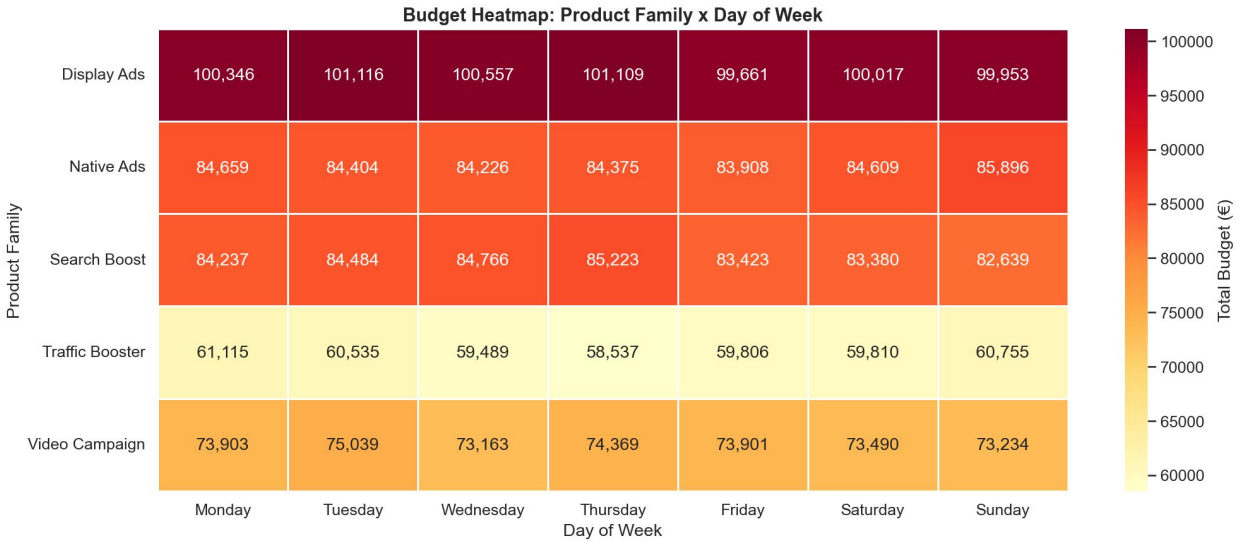
Weekday	Net Budget	Frequency	Av. Daily Budget	Media Budget
Monday	€ 404,260	1489	€ 271.5	€ 289.5
Tuesday	€ 405,577	1492	€ 271.8	€ 292.6
Wednesday	€ 402,201	1481	€ 271.5	€ 289.5
Thursday	€ 403,612	1484	€ 271.9	€ 292.6
Friday	€ 400,699	1478	€ 271.1	€ 289.5
Saturday	€ 401,306	1477	€ 271.7	€ 295.7
Sunday	€ 402,477	1479	€ 272.1	€ 295.7

Test	Comparison	Statistic	P-Value	Significance	Interpretation
Independent t-test	Weekend vs. Weekdays	t = -0.099373	p= 0.92084	No	No Significant difference

- Budget is uniformly distributed across 7 days.
- There is also no significant difference between Mean and media per day budget.
- Significance test tells there is no meaningful difference between weekends and weekdays.
- Product family mixed strategies matter for campaigns.

5. Dashboard Visuals and Tables

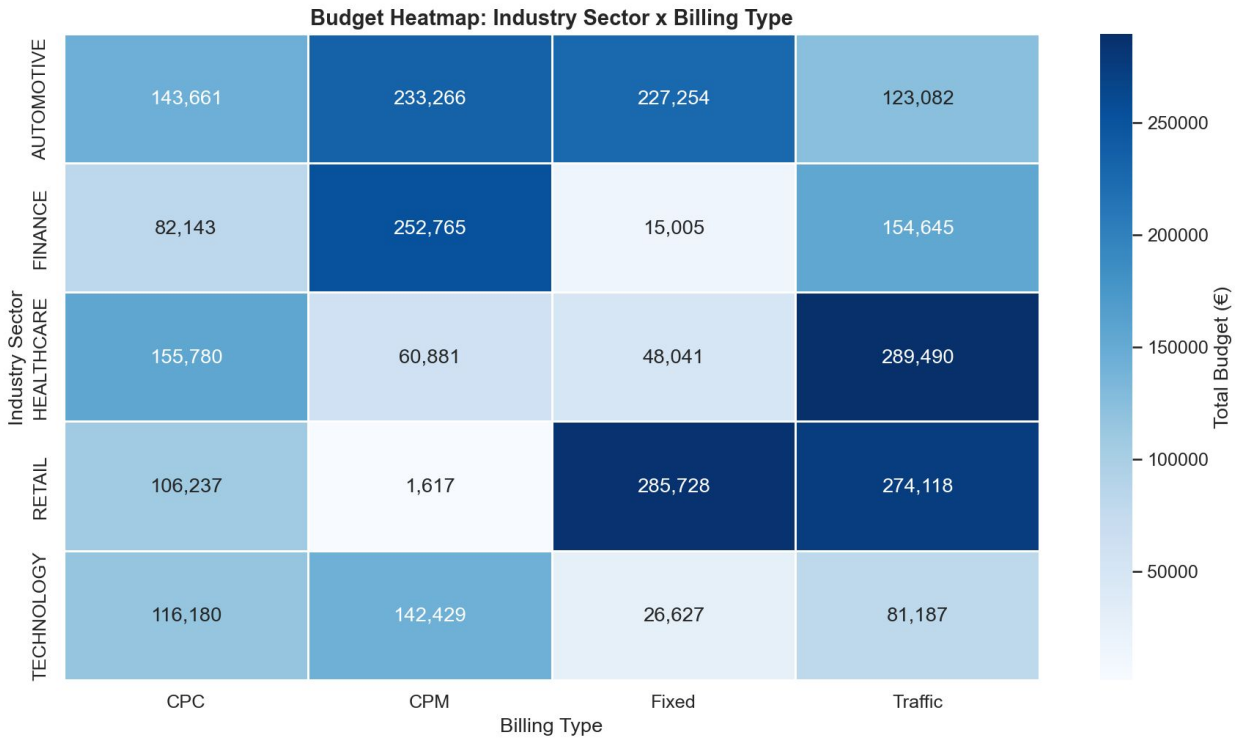
Visual: 14 (Product Family and Weekdays Budget Heatmap)



- The insignificance from weekdays effect on budget is also evident in this heatmap.
- The total budget over day of week for any product family remains the same for that product.
- Display Ads category leads the budget as previously shown in Visual: 7.

5. Dashboard Visuals and Tables

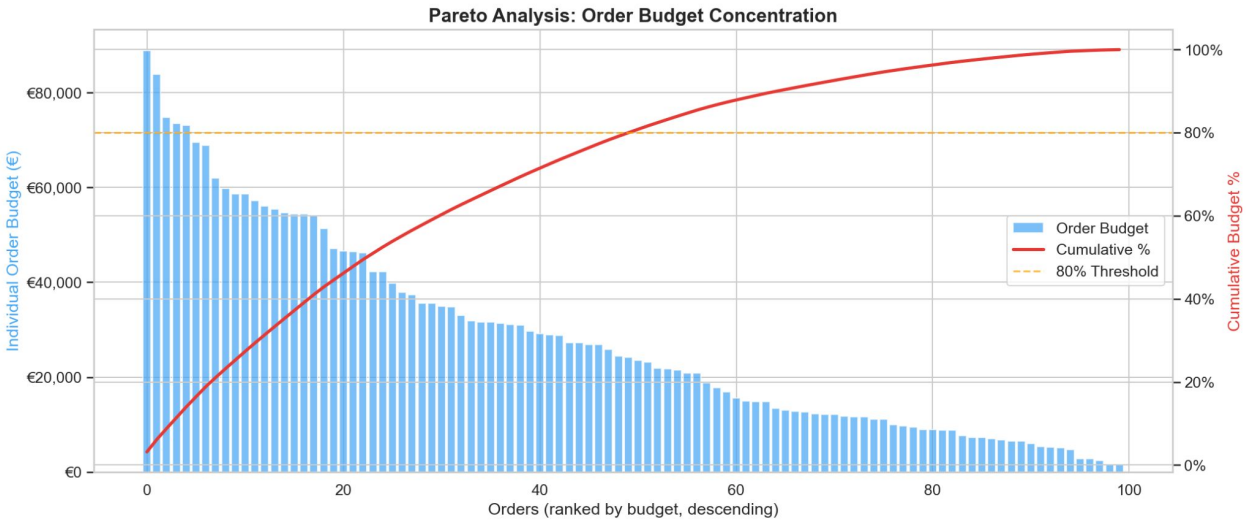
Visual: 15 (Billing Type and Industry Sector Budget Heatmap)



- The Automotive industry dominates.
- Billing type Traffic accounts for largest budget share.
- Automotive has two combinations with CPM and Fixed above €:200k.
- Retail has combinations with Fixed and Traffic above €:200k.
- Finance and Healthcare have one with CPM and Traffic respectively above €:200k.

5. Dashboard Visuals and Tables

Visual: 16 (Pareto Curve and Order Budget Concentration Analysis)



- 80/20 Pareto doesn't hold.
- 80% of the budget is concentrated among top-50 Orders.
- 20% of the budget is concentrated among the bottom-50 Orders.
- Top-10 Orders account for 25% of the budget.
- Top-20 Orders account for 44% of the budget.

6. Final Recommendations

Priority	CRITICAL	HIGH	HIGH	MEDIUM	MEDIUM	LOW
Recommendation	Start Q1-2026 campaign renewal. 69 Campaigns expired with no replacements.	Reduce Porsche dependency from 25% to 20%. Diversify the portfolio.	New Strategy for Tier-C under indexed customers. Growth plan for Audi. 15% share requires €115k investment.	18 long duration low budget orders have to be evaluated with cost benefit analysis.	Scale Native Ads: Highest average budget daily €115 and shortest duration. High expected returns.	Clean the data 'UNKNOWN'. It impairs the tracking.
Possible Owner	SALES	ACCOUNTS & STRATEGY	ACCOUNTS & MANAGEMENT	OPERATIONS	PRODUCT	DATA & TRACKING
Timeline	From Jan 2026	Q2 and Q3 2026	Q2 and Q3 2026	Q1 2026	Q2 2026	Q1 2026
Estimated Impact	At RR of €15,196/day, €2.5 Million of Annual Gap fillable.	~€140k Reallocation	~€115k Investment	Cost Reduction	Commission Revenue uplift	Data Quality

7. Methodology and Assumptions

HEADLINE FINDINGS:

- Dataset used: **synthetic_daily_revenue.csv**
- All analyses performed using: **Python 3.14**
- Imported Packages: **(datetime, numpy, pandas, scipy, matplotlib, seaborn)**
- Statistical significance assessed at: **ALPHA= 0.05**
- Outlier detection with **IQR** method with: **IQR factor = 1.5**
- Customer segmentation with **ABC: (A>20%, B>=15%, C<15%)**
- Detailed **Deep Dive Analysis** is available in the attached **Jupyter notebook** webpage (html format). **(F11_Analysis_Notebook.html)**
- Time series decomposition done with 30 days rolling window for seasonality and trend extraction.
- Dashboarding performed in **PowerBI** *free version*. Attached is the **dashboard file. (F11-task-PBI-Dashboard.pbix)**
- Due to limitation of free visuals, certain visuals were visualized in **Jupyter Notebook. (F11_Analysis_Notebook.ipynb)**
- Independent t-test used for Weekend vs. Weekday budget distribution.
- Kruskal-Wallis test used for Billing types and product families as significant difference in median and mean budgets.

ASSUMPTIONS:

- All amounts are assumed to be in **Euros**.
- **Fixed daily cap** model in budget assignment.
- **Pacing** is assumed to be **1.0** (100% daily budget is consumed everyday).
- No Revenue outcome data is available for ROI.